

# Reciprocal Measurement and the Mapping Origin of $c$

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**Version:** 1.6.0 (Revised: Dissipation-Compensation and Möbius-Residue Alignment)

## I. Physical Essence of Perception: Reciprocal Measurement

Perception is a physical process of reciprocal measurement embedded within the dissipation-compensation causal network. It may also be regarded as an emergent outcome generated by the direct topologically-correlated evolution of two topological manifolds.

- **Causal-Topological Instantiation:** Two quantum-evidential causal nodes undergo asynchronous evolution. Their causal intersection yields a Möbius-type topological residual manifold, namely the light residual  $\Psi_{light}$ . Only after passing joint evaluation of informational dissipation and network compensation cost can this interaction be instantiated and projected onto the physical rendering layer, becoming a “physically perceptible event”.
- **Instantiation Threshold (Emergent Ontological Ultraviolet Boundary):** An interaction must satisfy both causal consistency conditions and dissipation-compensation budget constraints. Only qualified interactions transition from the pending-for-instantiation state in causal space into observable physical reality. The Planck scale marks this ontological instantiation boundary rather than the intrinsic granular size of underlying causal nodes.

## II. Mapping the Measurement Limit: Emergent Origin of $c$

The speed-of-light constant  $c$  is not merely a kinematic constant. It is the **emergent causal upper bound of propagation** generated by the dissipation-compensation duality framework in the course of information-to-physical mapping. Space itself is no pre-existing background and is likewise an emergent product of the causal network. Spatial metrics and light-mediated reciprocal measurement originate from the same source and co-generate under distinct topological-rank regimes governed by the BBP spectral phase transition. Precisely this intrinsic coupling logically guarantees the invariance of locally measured speed of light.

- **Effective Routing Bandwidth  $c_{eff}$ :** In the high-dissipation primordial-universe regime ( $z \geq 4.1605$ ), substantial topological compensation overhead consumes the network’s computational resources, and the effective throughput for packet propagation  $c_{eff}$  decreases. Conformal gauge transformation rescales the emergent metric tensor synchronously; therefore for local observers within the rendering layer, the measured speed of light remains Lorentz-invariant.
- **Global Constraint from the Instantiation Boundary:** The emergent ontological ultraviolet boundary at the Planck scale sets a hard lower bound for logical-operation costs. If the compensation cost required for one reciprocal-measurement exceeds the network’s stability capacity, the process remains in the pending-for-instantiation state within causal space and cannot be rendered into observable spacetime events.

### III. Time: Cumulative Book-Keeping of Dissipation-Compensation Behaviours

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Time possesses no pre-existing axis-like container ontology; it emerges from cumulative book-keeping of numerous reciprocal-measurement events. Successive steps in the evolution of the world necessarily carry informational differences. Perception of the duration of such differences constitutes time. If no distinguishable informational difference exists between adjacent evolutionary steps, no new temporal increment will be generated and time tends to stagnate.

- **Non-Axis Nature:** Time is not a pre-existing background container. Under the constraint of the emergent causal upper bound of propagation, it is the sequential counting of successive topological dissipation-compensation operations.
- **Perception as Duration:** Every reciprocal-measurement interaction mediated by the Möbius light-residual manifold incurs irreducible topological routing overhead. These measurement events, weighted by compensation cost, stack continuously and manifest macroscopically as the experiential flow of temporal duration.

### IV. Logical Invariance of Locally-Measured Speed of Light across Reference Frames

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The locally measured speed of light remains invariant in all reference frames. Observers, observed systems and measurement apparatuses are all emergent products mapped from the same causal network, collectively governed by dissipation-compensation duality.

- **Mapping-Protocol Invariance:** All entities within the physical rendering layer share the same emergent ontological ultraviolet boundary and causal-routing constraints. Relative motion modulates the local effective throughput  $c_{eff}$ , yet conformal metric rescaling performs algebraic cancellation for every local observer and keeps the numerical value of measured speed of light invariant.

### Notes on Relationship with Earlier Versions

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Version 1.6.0 supersedes Version 1.5.2.

- Version 1.5.2 is an early operational sketch that describes the “hardware-interface” behaviour of the causal network under the simplified assumption of global fixed baud-rate synchronisation.
  - Version 1.6.0 integrates dissipation-compensation duality, the Möbius topological residual of light, BBP spectral rank transition and conformal-gauge covariance from the main SRE v6.0 paper. Reciprocal measurement is reinterpreted as an exchange process mediated by “light as topological residual”, constrained by dynamic topological-compensation costs instead of the limits of fixed single-logical-cycle synchronisation. As the operational-measurement layer, this version is fully compatible with SRE Axioms v1.6.
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